

test 1

Subject: Mathematics | Chapter: Algebra | Difficulty: medium

Question 1: If the roots of the quadratic equation $2x^2 - 5x + k = 0$ are $\frac{1}{2}$ and $\frac{1}{3}$, find the value of k .

Options:

- A) 3
- B) 2
- C) 6
- D) 1

Question 2: A rectangular garden has a length that is 3 meters more than its width. If the area of the garden is 70 square meters, what is the width of the garden?

Options:

- A) 7 meters
- B) 10 meters
- C) 5 meters
- D) 14 meters

Question 3: Solve the inequality $|2x - 1| < 5$.

Options:

- A) $-2 < x < 3$
- B) $x < -2$ or $x > 3$
- C) $x < 3$
- D) $x > -2$

Question 4: Find the sum of the infinite geometric series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$

Options:

- A) 1
- B) 2
- C) 3
- D) $\frac{1}{2}$

Question 5: A company's profit (P) in thousands of dollars is modeled by the function $P(x) = -x^2 + 10x - 16$, where x is the number of units produced in hundreds. How many units should be produced to maximize profit?

Options:

- A) 500 units
- B) 100 units
- C) 1000 units
- D) 200 units

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